

Digital Forensics Investigation

Disk Image Verification, Artifact Analysis and Timeline Reconstruction

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Portfolio Case Study - Detailed Project Oversight and Completion Record

All offensive-security activities described in this document were performed in authorized, isolated academic laboratory environments.

1. Project Overview

A comprehensive academic digital-forensics investigation using FTK Imager and Autopsy to verify forensic images, examine file systems and metadata, recover deleted artifacts, analyze communications and multimedia, and reconstruct an evidence-based timeline.

Project Field	Details
Role and Responsibility	Forensic examiner, evidence analyst, timeline analyst, and report author.
Environment and Tools	AccessData FTK Imager 4.7.1.2, Autopsy 4.22.1, E01 forensic images, Windows analysis workstation.
Completion Status	Completed - documented, tested, and included in the master portfolio.
Portfolio Classification	Academic / personal technical project with recruiter-facing case-study documentation.

2. Project Objectives and Success Criteria

- Verify evidence integrity using cryptographic hashes.
- Document chain-of-custody observations and acquisition metadata.
- Identify suspicious, encrypted, deleted, and relevant files.
- Analyze email, communications, metadata, and multimedia artifacts.
- Create a timeline and correlate evidence across multiple images.
- Produce a professional forensic report with findings and recommendations.

3. Project Oversight and Execution Process

3.1 Planning and Scope

Defined the authorized environment, project goal, tools, expected outputs, and safety limitations.

3.2 Implementation

Executed the technical tasks and maintained notes, commands, screenshots, and configuration evidence.

3.3 Validation and Testing

Compared actual output with objectives, confirmed repeatable behavior, and collected proof of completion.

3.4 Analysis and Reporting

Converted technical evidence into findings, defensive interpretation, limitations, and recommendations.

3.5 Portfolio Finalization

Prepared a recruiter-friendly summary, detailed case study, screenshots, source report, and public-release guidance.

4. Technical Methodology

1. Loaded forensic images and compared computed MD5/SHA-1 values with acquisition records.
2. Examined file-system structures, volume information, deleted files, and unallocated space.
3. Performed keyword searches, metadata extraction, and artifact bookmarking.
4. Reviewed email stores, contacts, media metadata, registry artifacts, and file histories.
5. Correlated timestamps and evidence sources into a structured investigative narrative.
6. Documented limitations, chain-of-custody concerns, and recommended follow-up actions.

5. Deliverables and Completion Evidence

Deliverable	Status	Evidence
Technical implementation / lab execution	Complete	Commands, configurations, code, or forensic workflow documented.
Testing and validation	Complete	Results and screenshots included below and in the original report.
Detailed project documentation	Complete	This case-study document and README.
Source/original evidence	Complete	Original report or source files stored in this project folder.
Recruiter-ready description	Complete	Summary and skills section included.
Public publishing review	Required before posting	Remove credentials, personal identifiers, confidential case data, and unnecessary target details.

6. Results and Outcomes

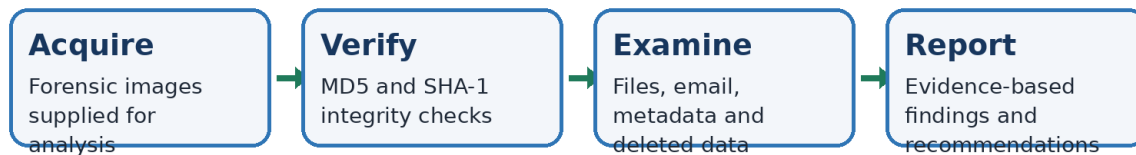
- All three images were reported as hash-verified, supporting evidence integrity.
- The examination recovered relevant documents, deleted files, communications, metadata, and multimedia artifacts.
- The final report demonstrates methodical evidence handling, correlation, and professional reporting.

7. Challenges, Decisions, and Lessons Learned

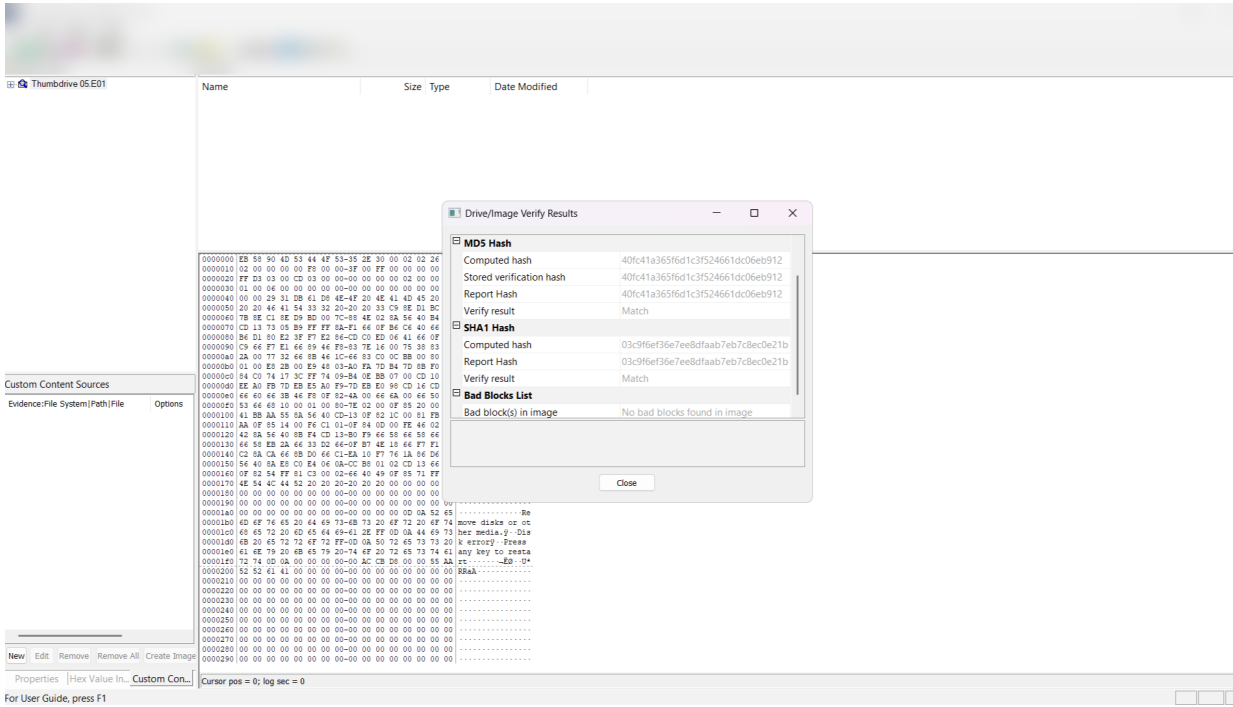
- Managing a large artifact set without overstating the significance of individual files.
- Normalizing timestamps and distinguishing direct evidence from inference.
- Preparing a public portfolio version without exposing confidential or case-specific details.

8. Screenshots and Supporting Evidence

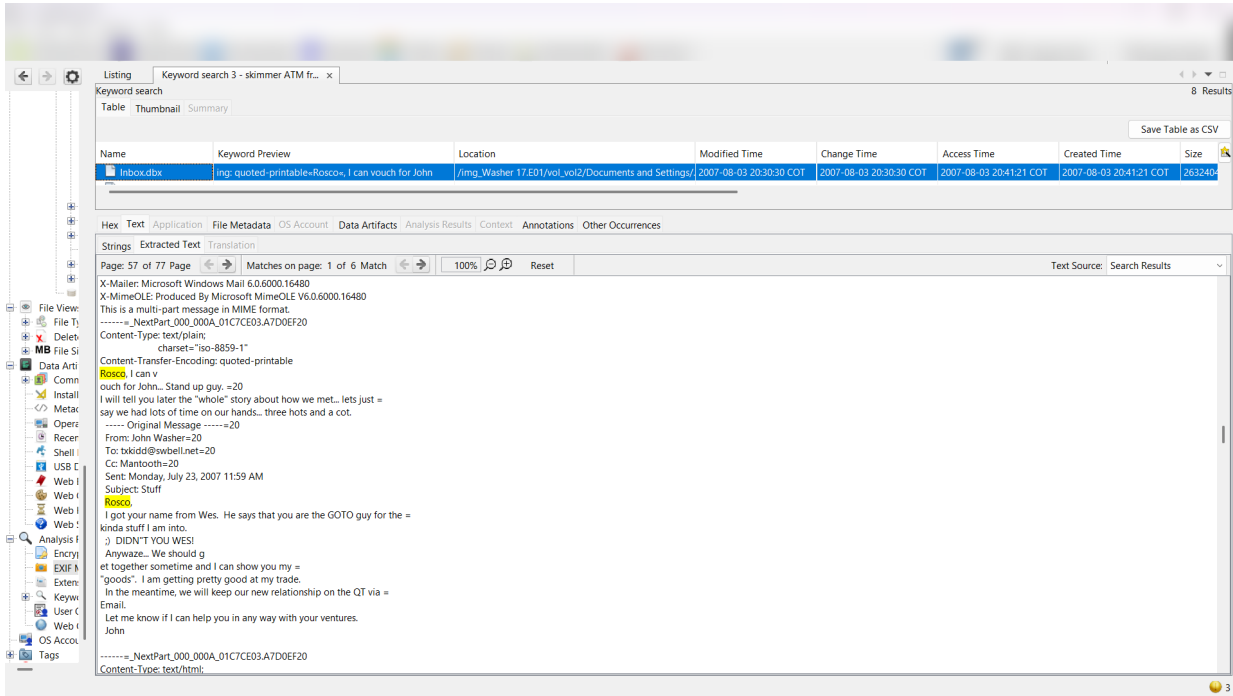
Digital Forensics Examination Workflow



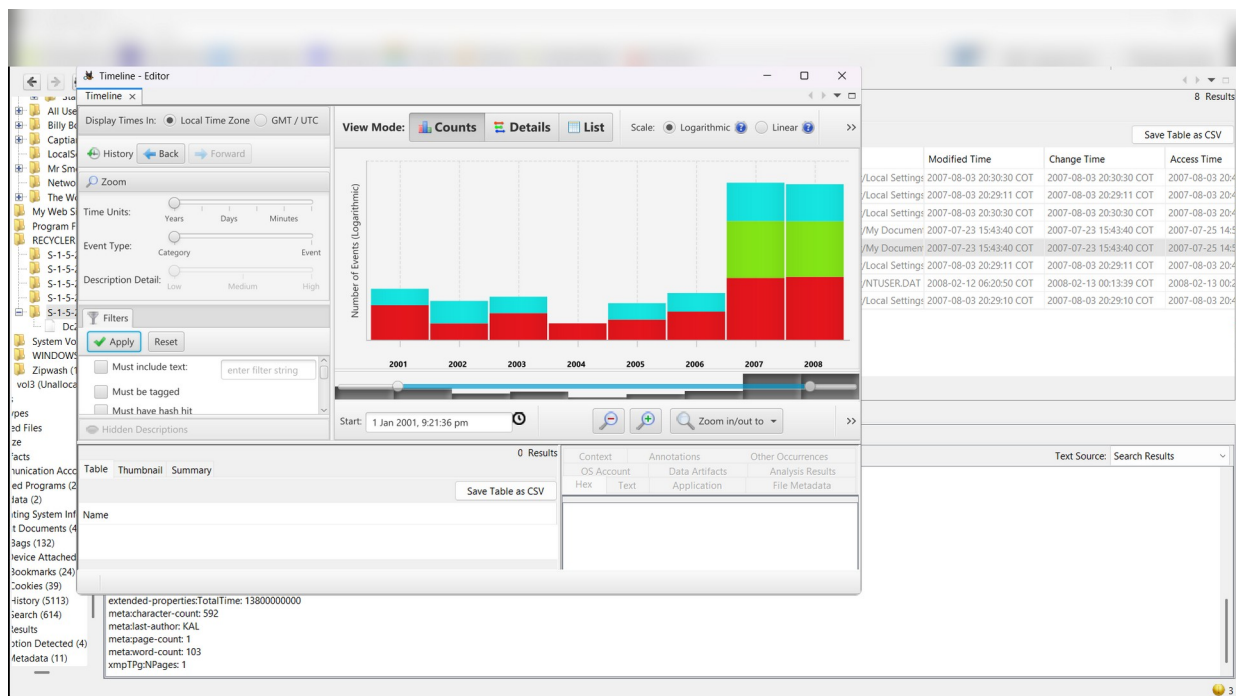
Forensic workflow diagram from acquisition verification through reporting.



Redacted evidence-integrity screenshot.



Redacted Autopsy analysis screenshot.



Redacted timeline-analysis screenshot.

9. Skills Demonstrated

FTK Imager, Autopsy, E01 analysis, hash verification, chain of custody, metadata, deleted-file recovery, email analysis, timeline reconstruction, forensic reporting.

10. Recommendations and Future Improvements

- Create a fully anonymized public edition with sensitive names and identifiers removed.
- Add an evidence matrix linking each conclusion to artifact IDs and screenshots.
- Export a machine-readable timeline and document repeatable search queries.

11. Portfolio and Resume Presentation

Digital Forensics Investigation - A comprehensive academic digital-forensics investigation using FTK Imager and Autopsy to verify forensic images, examine file systems and metadata, recover deleted artifacts, analyze communications and multimedia, and reconstruct an evidence-based timeline.

Recommended repository contents: README.md, Documentation/, Evidence/, Source/ or Original_Report/, and a clear authorization/privacy notice.

12. Completion Statement

This project is considered complete for portfolio use because the scope, implementation process, testing evidence, results, skills, limitations, and future improvements are documented. Before public release, sensitive data must be reviewed and removed.